KAPASA MAKASA UNIVERSITY

DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

BACHELOR OF SCIENCE IN CYBER SECURITY

CYS 425 – Mobile Device and Security and Forensic

Date: 23rd May, 2025 **Duration: 1 hour 25 Minutes**

Instructions: Answer all questions **Mark:** 35

1. What is the significance of "Provider cooperation" in cloud-based evidence retrieval?[2]
2. A forensic investigator is examining a suspect's mobile phone and finds that most documents were stored in Google Drive.
    a. What three (3) steps should they take to retrieve this data legally and effectively? [3]
3. What is file carving, and how is it used in mobile forensics? [2]
4. Explain the difference between logical recovery and physical recovery. [2]
5. What is unallocated space, and why is it important in forensic recovery? [2]
6. A banking app is suspected of leaking sensitive data. Describe how dynamic analysis could help identify malicious behaviour. [2]
7. A suspect stores incriminating files in iCloud. Outline three (3) challenges and legal steps required to retrieve this data. [2]
8. An investigator is pressured to alter findings to support a prosecution. Describe four (4) ethical principles that should guide their response? [4]
9. What are cache files, and how can they be useful in reconstructing a user's activities on a mobile device? [3]
10. What is the difference between network logs and device metadata? [2]
11. A hacker uses a mobile app to exfiltrate data via Bluetooth. What forensic evidence might remain on the device? [2]
12. What kind of evidence can be extracted from Call Detail Records (CDRs) in a criminal investigation? [1]
13. What is the difference between static analysis and dynamic analysis in reverse engineering? [2]
14. Compare the forensic value of GSM call logs vs. LTE session logs. Which provides more detailed evidence, and why? [4]
15. Why might a forensic investigator need to analyze both cellular and Wi-Fi logs in a single case? [2]

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mobile device and Security and forensic

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introduction to risk

(1.) provider cooperation this is an entity that controls cloud services and its significance in evidence retrieval is by allowing or giving access to investigators by not hindering them to cloud based evidence

(3) File carving is a recovery technique that involves recovering deleted data based on file signatures like the header and footer of a file in an unallocated space. Its used to recover data which is deleted eg deleted pictures which have a signature of .PNG to be covered based on PNG signature.

(4) Logical recovery is a technique that uses reference data location to recover files or data while physical recovery is a technique that involves getting a physical medium storage such as NAND memory storage and retrive data from there especially if logical recovery technic and file carving fails and the data is retrieved directly from the file structure or systems files of the memory.

(5) Unallocated space is just unassigned space or memory in the file system or storage and this is where deleted data is mostly stored before its overwritten.
Unallocated space is important in forensics recovery because it contains recently deleted data that can be used as evidence in crimes which was deleted in crimes as long as its not yet overwritten.

(6) Using dynamic analysis to identify the malicious behaviour, the apps will be run or executed in a controlled environment to see how the app runs its and its behaviour so as to identify the vulnerability in the app execution, during dynamic a tool such as Cobalt strike tool can be used.

~~(7)~~

(9) Cache files are temporary files of apps In mobile phones and their role is to save the usages of apps such as permissions, activities, locations accessed by and data used by used or accessed by an app and they are useful users activities by checking the settings in the cache files, permissions and other data that can then be used to understand what kind of actions or purpose the app was used for on a mobile device.

(10) Network logs are logs such as ~~ip and~~ location, 1, protocols used by a mobile ~~device~~ device, the network towers it connected to, for example IP addresses, MAC addresses data transmitted among others while device meta data is ~~info~~ information about the device such as the time ~~stamp~~ stamp, ~~location~~ activities used and even a version of the device.

(11) During extraction via bluetooth, ~~bluetooth~~ bluetooth logs remain such as information the mobile device paired with, its time stamp ~~at~~ and location.

(12) Call Detail Records evidence are call logs, messages and voice calls.

(13) Static analysis ~~is~~ is technique of analysing malware without running or executing them ~~but~~ but by observing the codes and file system and file contained by the malware while dynamic analysis is done by running or executing the malware in a controlled environment ~~on~~ to ~~under~~ understand its behaviours and intentions as it runs.

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(14) GSM call logs

incoming calls
outgoing calls
time stamp for calls
2nd generation
tower connected

LTE Session logs
Voice over internet
~~incoming calls~~
Enhance location tracking
time stamp
faster data transmission
for video calls and voice calls.
4th generation
Tower connected to
incoming calls, outgoing

The one that provides more ~~details~~ evidence is LTE session logs ~~because~~ especially when it its tracking location because it's shows the ~~geo~~ geographical area where the mobile device was ~~at~~ where it connected to ~~at~~ and the time stamp.

(15) The investigator needs to analyse both the cellular and wifi logs so that he can get detailed information of evidence from both side because the wifi logs can have information ~~which~~ which the cellular network cannot have like location, time stamp and vise versa and also both ~~evidence can information~~ information can give assurance and how accuracy the information is if the all align from both logs.